AI · EDUCATION · CAREER

AI Career Navigator

An Intelligent AI-Powered Career Guidance and Skill Recommendation System

Presented by: [Your Name] · B.Tech Artificial Intelligence and Machine Learning — 2nd Year · [Your College Name]



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THE CHALLENGE

The Career Confusion Challenge

Career opportunities

Learning resources

Career roadmap

- Students often have limited awareness of career opportunities after graduation.
- Job requirements, learning resources, and certifications are difficult to connect.
- Academic qualifications do not automatically translate into a clear career roadmap.
- Target users: Degree, B.Tech, MCA, and MBA students.
- Core problem: students need personalized, accessible, practical guidance based on education, skills, and goals.



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THE PLATFORM

Introducing AI Career Navigator

An intelligent platform that analyzes educational background, skills, interests, and career goals to recommend relevant opportunities and personalized learning plans.

From career confusion to a clear, personalized action plan.

- AI career chatbot
- Personalized career recommendations
- Skill-gap analysis
- Career roadmap generation
- Learning resources plus resume and portfolio guidance



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ONE PLATFORM · MANY PATHS

Designed for Every Student’s Career Journey

Recommendations depend on qualifications, skills, interests, and role requirements.



Degree

Software development, data analytics, business analysis, digital marketing, and pathways shaped by specialization and skills.



B.Tech

AI/ML engineering, software development, data engineering, and cloud computing.



MCA

Software engineering, AI/ML, data science, application development, and IT careers.



MBA

Business analytics, finance, marketing, human resources, and product management.

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THE TOOLKIT

Intelligent Features That Make a Difference

Each module turns a student question into a practical next step.



AI Career Chatbot

Natural-language answers to career questions.



Smart Career Matcher

Transparent matching logic based on the student profile.



Skill Gap Analyzer

Compares current skills with a selected career’s requirements.



Personalized Roadmap

Turns missing skills into stages, milestones, and projects.



Career Explorer

Shows roles, qualifications, and required skills.



Resume and Project Advisor

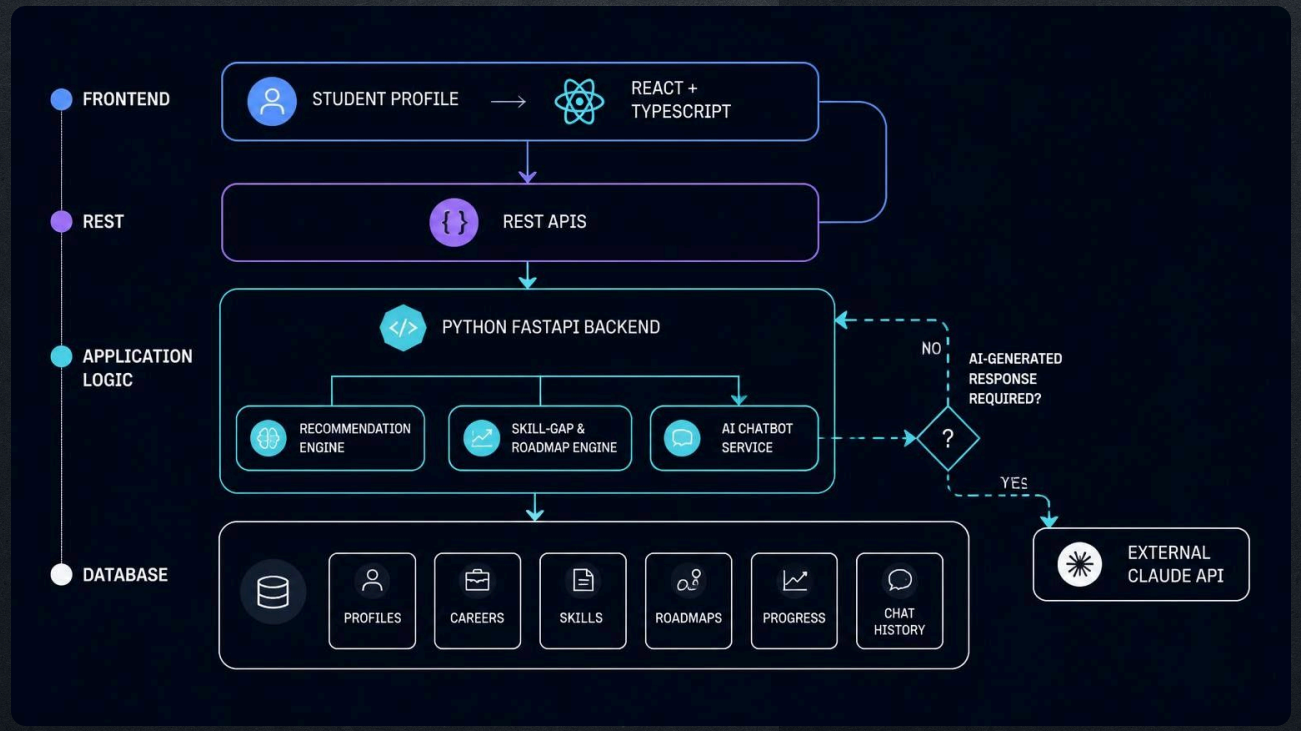
Supports portfolio and internship preparation.

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TECHNOLOGY ARCHITECTURE

How AI Career Navigator Works

A layered AI system connects student data, application logic, and personalized output.



Stack: React, TypeScript, Tailwind CSS · Python, FastAPI, Pydantic · SQLAlchemy · SQLite prototype with PostgreSQL as a future option · Git and

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The Intelligence Behind the System

Application logic handles matching and gap calculations; Claude assists with natural-language understanding when required.

Rule-based career matching

Qualification, interest, skill, and eligibility matching with explainable recommendations.

Claude API integration

Understands natural-language questions and assists with explanations and learning plans.

Skill-gap analysis

Compares current and required skills, then prioritizes learning tasks.

Structured career knowledge

Stores career descriptions, qualifications, skills, and resources for consistent retrieval.



Student profile



Career matching



Skill-gap identification



AI-assisted roadmap



Actionable recommendations

The initial version uses an existing AI model through an API plus a custom recommendation engine; it does not train a large language model from scratch.

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ILLUSTRATIVE WORKFLOW · NOT A REAL TEST RESULT

From Student Profile to Career Roadmap

1. **Create profile:** B.Tech in AI/ML, second year, basic Python and introductory programming, 8 hours per week.
2. **Select target:** Machine Learning Engineer.
3. **Compare skills:** current skills against role requirements.
4. **Identify priorities:** mathematics, Python, data handling, machine learning fundamentals, and model evaluation.
5. **Receive resources and projects:** learning resources and projects.
6. **Track staged roadmap:** follow staged progress toward the target.

Example projects: student performance prediction · house price prediction · customer segmentation · model deployment.



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Building a Reliable and Useful Career Guidance Platform

Implementation

Requirements and architecture → backend and database → matching and skill-gap logic → Claude integration → frontend dashboard → integration and testing → deployment and documentation.

Expected outcomes

Easier access to career information, more personalized learning plans, clearer job-skill requirements, and organized career preparation.

Testing checklist

Validate inputs, matching rules, gap calculations, chatbot failures, database persistence, responsive layouts, relevance, and explainability.

Limitations

AI responses may be inaccurate; job-market information needs current sources; results depend on data quality and user inputs; API usage may incur costs.

No measurable improvement is claimed without evaluation.

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THE NEXT DECISION IS YOURS

Empowering Students to Make Informed Career Decisions

AI Career Navigator combines career-matching logic, skill-gap analysis, structured learning roadmaps, and AI-powered conversation to make career exploration more personalized, accessible, and actionable.

Your Education. Your Skills. Your Future. One Intelligent Career Navigator.

Future scope

Verified live jobs and internships · resume analysis · semantic search and Retrieval-Augmented Generation · regional and international pathways · trusted learning-platform integrations · multilingual guidance · progress-based recommendations · evaluated user feedback.

Thank You

Questions & Answers

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